

1. Calcule, se existirem, os seguintes limites:

- |                                                                                    |                                                                                          |                                                                              |
|------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|------------------------------------------------------------------------------|
| a) $\lim_{x \rightarrow \frac{1}{2}^-} \frac{\ln(1-2x)}{\operatorname{tg}(\pi x)}$ | b) $\lim_{x \rightarrow 0^+} \frac{\ln x}{\cotg x}$                                      | c) $\lim_{x \rightarrow 1^+} (\ln x)^{(x-1)}$                                |
| d) $\lim_{x \rightarrow +\infty} \frac{\ln x}{e^{2x}}$                             | e) $\lim_{x \rightarrow +\infty} \frac{x e^x}{e^{x^2}}$                                  | f) $\lim_{x \rightarrow 0^+} x^p \ln x, p > 0$                               |
| g) $\lim_{x \rightarrow 0^+} x^{\operatorname{tg}(x^2)}$                           | h) $\lim_{x \rightarrow 0} \left( \frac{1}{\ln(1+x)} - \frac{1}{e^x - 1} \right)$        | i) $\lim_{x \rightarrow 0^+} (x \operatorname{sen} x)^{\operatorname{tg} x}$ |
| j) $\lim_{x \rightarrow 0} (e^x + 3x)^{\frac{1}{x}}$                               | k) $\lim_{x \rightarrow 0} \left( \frac{1}{1 - \cos x} - \frac{2}{x^2} \right)$          | l) $\lim_{x \rightarrow 0^+} \left( \frac{1}{x} + \ln x \right)$             |
| m) $\lim_{x \rightarrow 0} \frac{\ln(1+x^2)}{x \arctg x}$                          | n) $\lim_{x \rightarrow \frac{\pi}{2}^+} (\operatorname{tg} x \sec x - \sec^2 x)$        | o) $\lim_{x \rightarrow 0} \frac{x \sin x + 2x^2}{e^x + e^{-x} - 2}$         |
| p) $\lim_{x \rightarrow 0} (1 + 5x)^{\frac{3}{x}}$                                 | q) $\lim_{x \rightarrow 0} (1 + \operatorname{sen} 2x)^{\frac{1}{\operatorname{sen} x}}$ | r) $\lim_{x \rightarrow 0^+} (\operatorname{sen} x)^{\frac{1}{\ln x}}$       |
| s) $\lim_{x \rightarrow 0^+} (1 + 3x)^{\frac{1}{\arctg(2x)}}$                      | t) $\lim_{x \rightarrow 0^+} (1 - \cos x)^x$                                             | u) $\lim_{x \rightarrow 1^-} (2 - x)^{\operatorname{tg}(\frac{\pi x}{2})}$   |